Week3&4_Malekshahi

January 12, 2026

[8]:
```
# White House Visitor Logs (Weeks 3 & 4 Exercise) - Visual Story (Python)
# Author: Shaghayegh Malekshahi
# Course: DSC 530 - Data Presentation
# Term: Winter 2026

# 1) IMPORTS
import os
import re
import glob
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
%matplotlib inline
```

[3]:
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# A) CONFIGURATION

EXTRACT_DIR = r"C:
 ↪\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter␣
 ↪2026\Data presentation\week 3&4\assignment"
OUTPUT_DIR  = r"C:
 ↪\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter␣
 ↪2026\Data presentation\week 3&4\assignment\output"

Path(OUTPUT_DIR).mkdir(parents=True, exist_ok=True)
Path(os.path.join(OUTPUT_DIR, "figures")).mkdir(parents=True, exist_ok=True)

# If True, uses all CSVs in the folder (slower but acceptable).
# If False, uses a curated list of 6 (>=5 required).
USE_ALL_DATASETS = False

DEFAULT_PICK = [
    "2022.01_WAVES-ACCESS-RECORDS.csv",
    "2022.06_WAVES-ACCESS-RECORDS.csv",
    "2022.12_WAVES-ACCESS-RECORDS.csv",
    "2023.03_WAVES-ACCESS-RECORDS.csv",
```

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    "2023.06_WAVES-ACCESS-RECORDS.csv",
    "2023.12_WAVES-ACCESS-RECORDS.csv",
]
```

[4]:
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# B) FILE DISCOVERY

all_csvs = sorted(glob.glob(os.path.join(EXTRACT_DIR, "*.csv")))
if not all_csvs:
    raise FileNotFoundError(f"No CSV files found in: {EXTRACT_DIR}")

if USE_ALL_DATASETS:
    selected_csvs = all_csvs
else:
    picks = [os.path.join(EXTRACT_DIR, fn) for fn in DEFAULT_PICK]
    selected_csvs = [p for p in picks if os.path.exists(p)]

    # If some picked months aren't present, fall back to first 6 found
    if len(selected_csvs) < 5:
        selected_csvs = all_csvs[:6]

print(f"Found {len(all_csvs)} total CSVs.")
print(f"Using {len(selected_csvs)} CSVs (>=5 required):")
for f in selected_csvs:
    print(" -", os.path.basename(f))
```

```
Found 25 total CSVs.
Using 6 CSVs (>=5 required):
 - 2022.01_WAVES-ACCESS-RECORDS.csv
 - 2022.06_WAVES-ACCESS-RECORDS.csv
 - 2022.12_WAVES-ACCESS-RECORDS.csv
 - 2023.03_WAVES-ACCESS-RECORDS.csv
 - 2023.06_WAVES-ACCESS-RECORDS.csv
 - 2023.12_WAVES-ACCESS-RECORDS.csv
```

[5]:
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# C) LOADING + COLUMN STANDARDIZATION

def normalize_col(c: str) -> str:
    c = c.strip()
    c = re.sub(r"[^\w]+", "_", c)
    c = re.sub(r"_+", "_", c).strip("_")
    return c.upper()

CANON_MAP = {
    "APPOINTMENT_START_DATE": "APPT_START_DATE",
    "APPOINTMENT_END_DATE": "APPT_END_DATE",
    "APPOINTMENT_MADE_DATE": "APPT_MADE_DATE",
    "APPOINTMENT_CANCEL_DATE": "APPT_CANCEL_DATE",
    "MEETING_LOCATION": "MEETING_LOC",
```

```python
    "LAST_ENTRY_DATE": "LASTENTRYDATE",
    "RELEASE_DATE": "RELEASEDATE",
}

KEEP_CANON = [
    "TOA",                  # arrival time (often best indicator of actual visit␣
 ↪time)
    "APPT_START_DATE",       # fallback if TOA missing
    "APPT_END_DATE",
    "APPT_MADE_DATE",
    "APPT_CANCEL_DATE",
    "TOTAL_PEOPLE",
    "MEETING_LOC",
    "MEETING_ROOM",
    "ACCESS_TYPE",
    "LASTENTRYDATE",         # fallback if TOA and APPT missing
    "RELEASEDATE",
]

def infer_month_from_filename(filename: str) -> str:
    m = re.search(r"(20\d{2})[._-](\d{2})", filename)
    if m:
        return f"{m.group(1)}-{m.group(2)}"
    y = re.search(r"(20\d{2})", filename)
    return f"{y.group(1)}-00" if y else "unknown"

frames = []
for fp in selected_csvs:
    d = pd.read_csv(fp, low_memory=False)

    # Normalize column names
    d.columns = [normalize_col(c) for c in d.columns]
    d = d.rename(columns={c: CANON_MAP.get(c, c) for c in d.columns})

    # Keep only useful columns that exist in this file
    keep = [c for c in KEEP_CANON if c in d.columns]
    dd = d[keep].copy()

    # Provenance
    dd["SOURCE_FILE"] = os.path.basename(fp)
    dd["SOURCE_MONTH_FILE"] = infer_month_from_filename(os.path.basename(fp))

    frames.append(dd)

df = pd.concat(frames, ignore_index=True)
print("Combined rows:", len(df))
```

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Combined rows: 392390
```

[6]:
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# D) CLEANING + DATETIME FEATURE CREATION

# Mixed date formats exist across files; format="mixed" handles both:
date_cols = ["TOA", "APPT_START_DATE", "APPT_END_DATE", "APPT_MADE_DATE",␣
↪"APPT_CANCEL_DATE", "LASTENTRYDATE", "RELEASEDATE"]
for c in date_cols:
    if c in df.columns:
        df[c] = pd.to_datetime(df[c], errors="coerce", format="mixed")

# TOTAL_PEOPLE numeric
if "TOTAL_PEOPLE" in df.columns:
    df["TOTAL_PEOPLE"] = pd.to_numeric(df["TOTAL_PEOPLE"], errors="coerce").
↪fillna(1)
else:
    df["TOTAL_PEOPLE"] = 1

# VISIT_DT: best available timestamp (TOA > APPT_START_DATE > LASTENTRYDATE)
df["VISIT_DT"] = df.get("TOA").fillna(df.get("APPT_START_DATE")).fillna(df.
↪get("LASTENTRYDATE"))

# Remove rows without a usable visit timestamp (documented filter)
df = df[df["VISIT_DT"].notna()].copy()

# Derived time fields
df["VISIT_DATE"]  = df["VISIT_DT"].dt.date
df["VISIT_MONTH"] = df["VISIT_DT"].dt.to_period("M").astype(str)
df["VISIT_DOW"]   = df["VISIT_DT"].dt.day_name()
df["VISIT_HOUR"]  = df["VISIT_DT"].dt.hour

# Save cleaned dataset
clean_path = os.path.join(OUTPUT_DIR, "cleaned_visits.csv")
df.to_csv(clean_path, index=False)
print("Saved cleaned dataset:", clean_path)
```

```
Saved cleaned dataset:
C:\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter
2026\Data presentation\week 3&4\assignment\output\cleaned_visits.csv
```

[7]:
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# E) SIX VISUALIZATIONS
# ========================
plt.rcParams["figure.dpi"] = 140

def save_fig(filename_no_ext: str) -> str:
    path = os.path.join(OUTPUT_DIR, "figures", f"{filename_no_ext}.png")
    plt.tight_layout()
    plt.savefig(path, bbox_inches="tight")
    plt.close()
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    return path

figure_paths = []

# 1) LINE CHART (required list): Daily visit volume
daily = df.groupby("VISIT_DATE").size().reset_index(name="visits")
daily["VISIT_DATE"] = pd.to_datetime(daily["VISIT_DATE"])
daily = daily.sort_values("VISIT_DATE")

plt.figure(figsize=(10,4))
plt.plot(daily["VISIT_DATE"], daily["visits"])
plt.title("Daily Visit Volume (Selected Months)")
plt.xlabel("Date")
plt.ylabel("Number of visits")
figure_paths.append(save_fig("01_line_daily_visits"))

# 2) STACKED BAR WITH TIME (required list): Monthly visits by meeting location␣
↪(Top 4 + Other)
loc = df["MEETING_LOC"].fillna("Unknown") if "MEETING_LOC" in df.columns else␣
↪pd.Series(["Unknown"] * len(df))
monthly_loc = df.assign(MEETING_LOC_C=loc).
↪groupby(["VISIT_MONTH","MEETING_LOC_C"]).size().reset_index(name="visits")

top_locs = (monthly_loc.groupby("MEETING_LOC_C")["visits"].sum()
            .sort_values(ascending=False).head(4).index.tolist())

monthly_loc["MEETING_LOC_C"] = monthly_loc["MEETING_LOC_C"].where(
    monthly_loc["MEETING_LOC_C"].isin(top_locs), "Other"
)

pivot_loc = (monthly_loc.groupby(["VISIT_MONTH","MEETING_LOC_C"])["visits"]
             .sum().unstack(fill_value=0).sort_index())

plt.figure(figsize=(10,4.5))
bottom = np.zeros(len(pivot_loc))
for col in pivot_loc.columns:
    plt.bar(pivot_loc.index, pivot_loc[col].values, bottom=bottom, label=col)
    bottom += pivot_loc[col].values

plt.title("Monthly Visits by Meeting Location (Top 4 + Other)")
plt.xlabel("Month")
plt.ylabel("Number of visits")
plt.xticks(rotation=45, ha="right")
plt.legend(loc="upper right", fontsize=8)
figure_paths.append(save_fig("02_stackedbar_monthly_location"))

# 3) BAR CHART (required list): Top 10 meeting rooms
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room = df["MEETING_ROOM"].fillna("Unknown") if "MEETING_ROOM" in df.columns␣
↪else pd.Series(["Unknown"] * len(df))
top_rooms = room.value_counts().head(10).sort_values()

plt.figure(figsize=(8,5))
plt.barh(top_rooms.index, top_rooms.values)
plt.title("Top 10 Meeting Rooms by Visit Count")
plt.xlabel("Number of visits")
plt.ylabel("Meeting room")
figure_paths.append(save_fig("03_bar_top_rooms"))

# 4) SCATTERPLOT WITH TIME (required list): Visit size over time (sampled)
sample = df.sample(min(6000, len(df)), random_state=42)

plt.figure(figsize=(10,4))
plt.scatter(sample["VISIT_DT"], sample["TOTAL_PEOPLE"], s=8, alpha=0.4)
plt.title("Visit Size Over Time (Sampled)")
plt.xlabel("Visit datetime")
plt.ylabel("Total people on record")
figure_paths.append(save_fig("04_scatter_time_total_people"))

# 5) STEP CHART (required list): Cumulative visits
daily_cum = daily.copy()
daily_cum["cum_visits"] = daily_cum["visits"].cumsum()

plt.figure(figsize=(10,4))
plt.step(daily_cum["VISIT_DATE"], daily_cum["cum_visits"], where="post")
plt.title("Cumulative Visits Over Time (Selected Months)")
plt.xlabel("Date")
plt.ylabel("Cumulative visits")
figure_paths.append(save_fig("05_step_cumulative_visits"))

# 6) COLUMN CHART (required list): Visits by day of week
dow_order =␣
↪["Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday"]
dow_counts = df["VISIT_DOW"].value_counts().reindex(dow_order)

plt.figure(figsize=(9,4))
plt.bar(dow_counts.index, dow_counts.values)
plt.title("Visits by Day of Week")
plt.xlabel("Day of week")
plt.ylabel("Number of visits")
plt.xticks(rotation=30, ha="right")
figure_paths.append(save_fig("06_column_dow"))

print("\nSaved figures:")
for p in figure_paths:
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    print(" -", p)
```

```
Saved figures:
 - C:\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter
2026\Data presentation\week
3&4\assignment\output\figures\01_line_daily_visits.png
 - C:\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter
2026\Data presentation\week
3&4\assignment\output\figures\02_stackedbar_monthly_location.png
 - C:\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter
2026\Data presentation\week 3&4\assignment\output\figures\03_bar_top_rooms.png
 - C:\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter
2026\Data presentation\week
3&4\assignment\output\figures\04_scatter_time_total_people.png
 - C:\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter
2026\Data presentation\week
3&4\assignment\output\figures\05_step_cumulative_visits.png
 - C:\Users\shagh\DSC530\envs\book_env1\Hands_On_Data_with_panda_edition2\winter
2026\Data presentation\week 3&4\assignment\output\figures\06_column_dow.png
```

[9]:
```
from IPython.display import display, Image

print("========================================")
print("JUPYTER OUTPUT VERIFICATION")
print("========================================\n")

# Show cleaned data preview
print("Cleaned dataset preview (first 10 rows):")
display(df.head(10))

print("\nDataset summary:")
print("Total rows after cleaning:", len(df))
print("Number of datasets used:", df['SOURCE_FILE'].nunique())

print("\nDatasets included:")
display(df['SOURCE_FILE'].value_counts())

# Display all saved figures inline
print("\nDisplaying saved figures:\n")
for p in figure_paths:
    display(Image(filename=p))
```

```
========================================
JUPYTER OUTPUT VERIFICATION
========================================

Cleaned dataset preview (first 10 rows):
```

```
                  TOA APPT_START_DATE       APPT_END_DATE      APPT_MADE_DATE  \
0 2022-01-01 12:22:00      2022-01-01 2022-01-01 11:59:00 2021-12-31 01:54:00
1                 NaT      2022-01-02 2022-01-02 11:59:00 2022-01-02 09:03:00
2 2022-01-02 10:59:00      2022-01-02 2022-01-02 11:59:00 2021-12-30 04:27:00
3 2022-01-02 14:01:00      2022-01-02 2022-01-02 11:59:00 2021-12-31 03:03:00
4                 NaT      2022-01-02 2022-01-02 11:59:00 2021-12-27 04:04:00
5                 NaT      2022-01-03 2022-01-03 11:59:00 2021-12-10 01:00:00
6 2022-01-03 12:52:00      2022-01-03 2022-01-03 11:59:00 2022-01-02 09:03:00
7 2022-01-03 12:48:00      2022-01-03 2022-01-03 11:59:00 2022-01-03 08:37:00
8                 NaT      2022-01-03 2022-01-03 11:59:00 2021-12-29 01:00:00
9                 NaT      2022-01-03 2022-01-03 11:59:00 2021-12-10 01:00:00

  APPT_CANCEL_DATE  TOTAL_PEOPLE MEETING_LOC MEETING_ROOM ACCESS_TYPE  \
0              NaT           1.0          WH       WW-132          VA
1              NaT           1.0          WH       WW G47          VA
2              NaT           1.0        OEOB           97          VA
3              NaT           1.0          WH         WHSR          VA
4              NaT           1.0        OEOB           97          VA
5              NaT           2.0        OEOB         412A          VA
6              NaT           1.0        OEOB          386          VA
7              NaT           1.0        OEOB   South Cour          VA
8              NaT           3.0        OEOB          226          VA
9              NaT           2.0        OEOB         412A          VA

        LASTENTRYDATE RELEASEDATE                      SOURCE_FILE  \
0 2021-12-31 01:54:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
1 2022-01-02 09:03:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
2 2021-12-30 04:27:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
3 2021-12-31 03:03:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
4 2021-12-27 04:04:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
5 2021-12-10 01:00:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
6 2022-01-02 09:03:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
7 2022-01-03 08:37:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
8 2021-12-29 01:00:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv
9 2021-12-10 01:00:00  2022-05-04  2022.01_WAVES-ACCESS-RECORDS.csv

  SOURCE_MONTH_FILE            VISIT_DT  VISIT_DATE VISIT_MONTH VISIT_DOW  \
0           2022-01 2022-01-01 12:22:00  2022-01-01     2022-01  Saturday
1           2022-01 2022-01-02 00:00:00  2022-01-02     2022-01    Sunday
2           2022-01 2022-01-02 10:59:00  2022-01-02     2022-01    Sunday
3           2022-01 2022-01-02 14:01:00  2022-01-02     2022-01    Sunday
4           2022-01 2022-01-02 00:00:00  2022-01-02     2022-01    Sunday
5           2022-01 2022-01-03 00:00:00  2022-01-03     2022-01    Monday
6           2022-01 2022-01-03 12:52:00  2022-01-03     2022-01    Monday
7           2022-01 2022-01-03 12:48:00  2022-01-03     2022-01    Monday
8           2022-01 2022-01-03 00:00:00  2022-01-03     2022-01    Monday
9           2022-01 2022-01-03 00:00:00  2022-01-03     2022-01    Monday
```

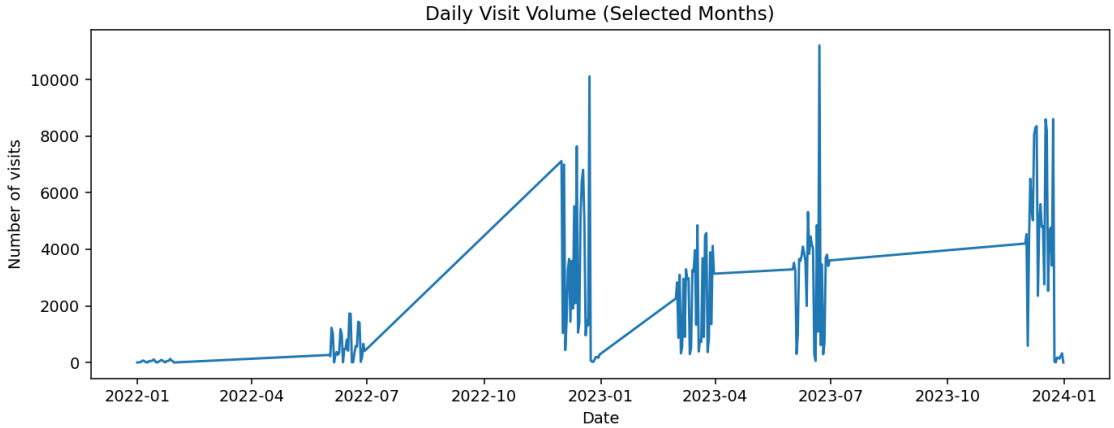
```
   VISIT_HOUR
0          12
1           0
2          10
3          14
4           0
5           0
6          12
7          12
8           0
9           0

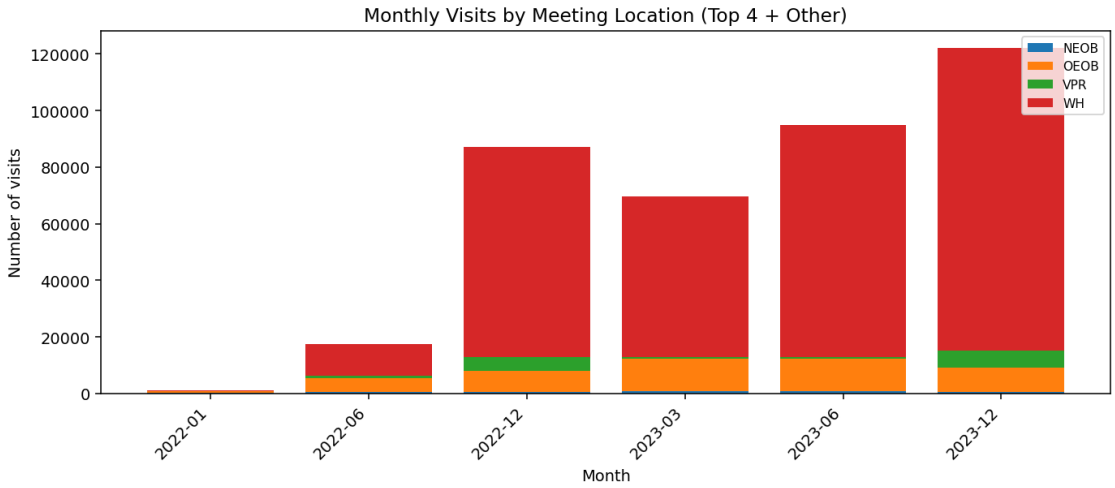

Dataset summary:
Total rows after cleaning: 392390
Number of datasets used: 6

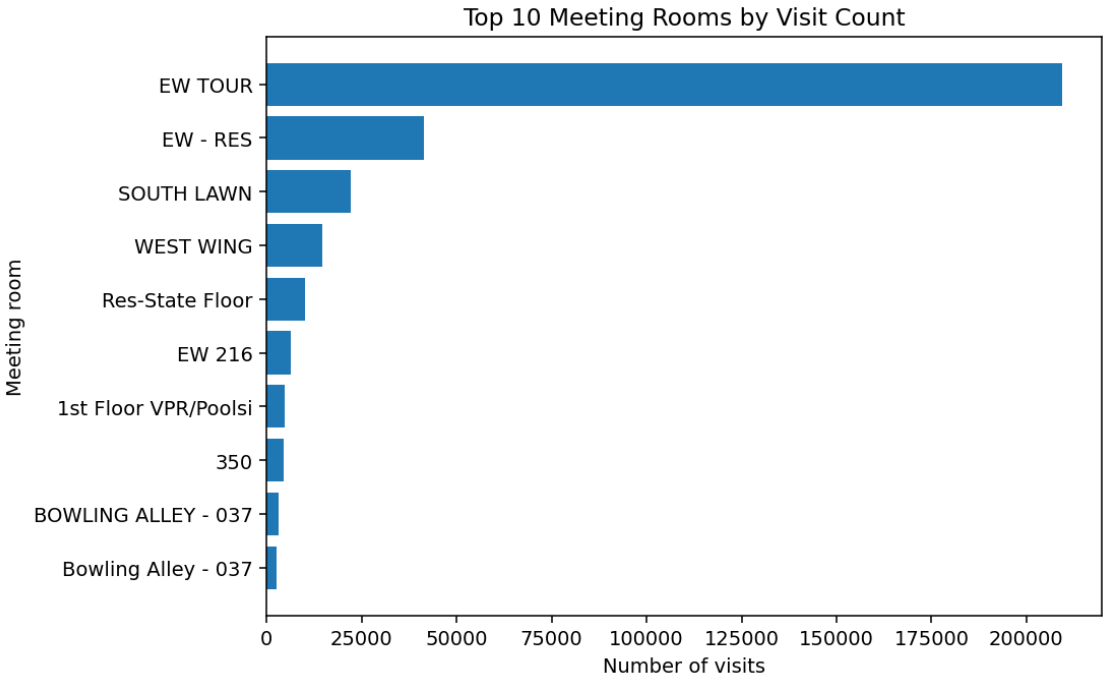
Datasets included:

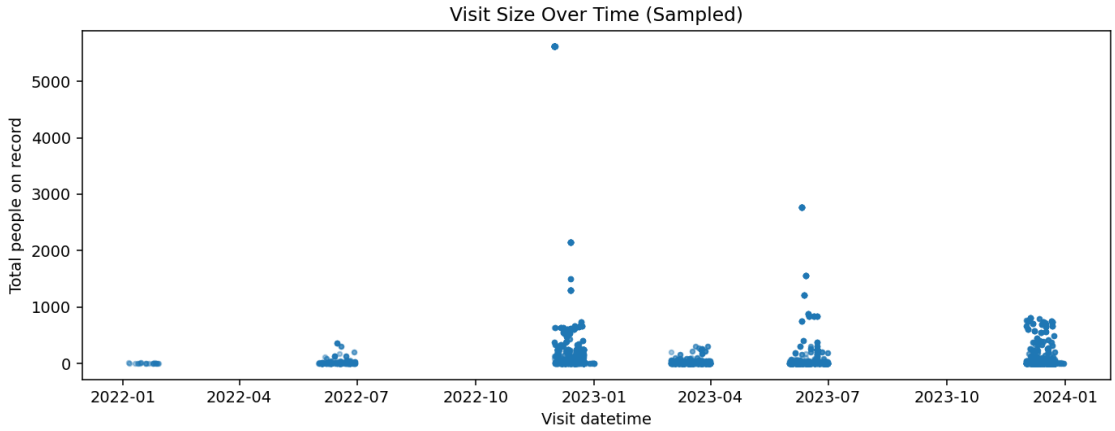
SOURCE_FILE
2023.12_WAVES-ACCESS-RECORDS.csv    122138
2023.06_WAVES-ACCESS-RECORDS.csv     94836
2022.12_WAVES-ACCESS-RECORDS.csv     87061
2023.03_WAVES-ACCESS-RECORDS.csv     69770
2022.06_WAVES-ACCESS-RECORDS.csv     17400
2022.01_WAVES-ACCESS-RECORDS.csv      1185
Name: count, dtype: int64

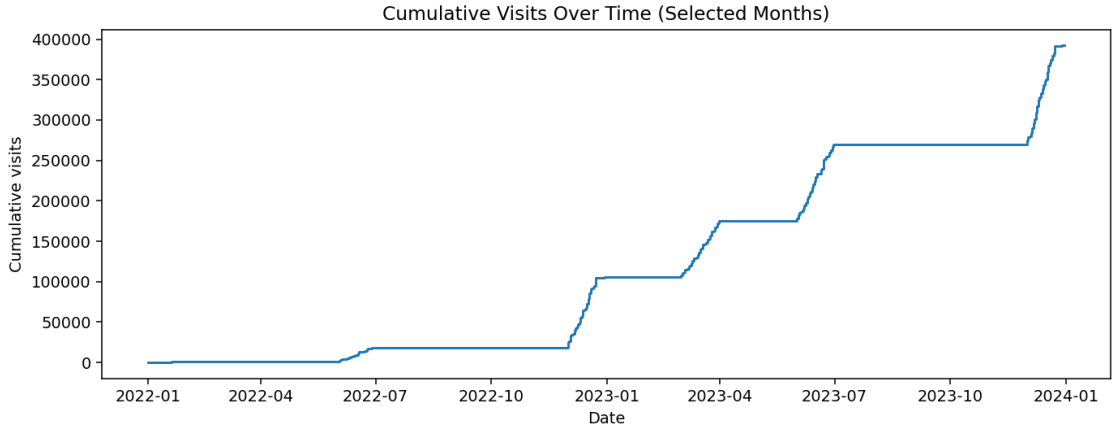

Displaying saved figures:
```

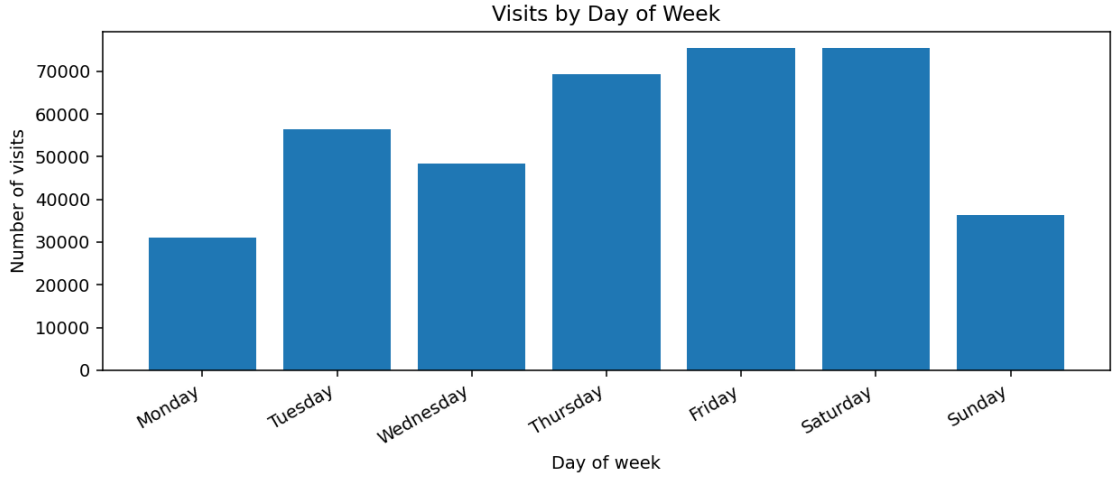



Monthly Visits by Meeting Location (Top 4 + Other)
NEOB
OEOB
VPR
WH
Number of visits
0
20000
40000
60000
80000
100000
120000
2022-01
2022-06
2022-12
2023-03
2023-06
2023-12
Month


Top 10 Meeting Rooms by Visit Count
Meeting room
EW TOUR
EW - RES
SOUTH LAWN
WEST WING
Res-State Floor
EW 216
1st Floor VPR/Poolsi
350
BOWLING ALLEY - 037
Bowling Alley - 037
0
25000
50000
75000
100000
125000
150000
175000
200000
Number of visits


Visit Size Over Time (Sampled)
Total people on record
5000
4000
3000
2000
1000
0
2022-01
2022-04
2022-07
2022-10
2023-01
2023-04
2023-07
2023-10
2024-01
Visit datetime


Cumulative Visits Over Time (Selected Months)
Cumulative visits
400000
350000
300000
250000
200000
150000
100000
50000
0
2022-01
2022-04
2022-07
2022-10
2023-01
2023-04
2023-07
2023-10
2024-01
Date


Visits by Day of Week
Number of visits
70000
60000
50000
40000
30000
20000
10000
0
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of week